

Materials Harness

Every test run to date, and what each one found

Generated 17 September 2026 · production engine MACE-MPA-0 medium (cpu/float32, settings tag c2480e74) · second engine MACE-MP-0 medium (cpu/float64, tag 207ccc81)

How to read this document

Ten tests and two publication audits are recorded here. The tests are not interchangeable: three measure accuracy against DFT, two measure whether the product's *decisions* are safe, one measures whether the pipeline finds the right structure at all, and the rest check data integrity and engine provenance. The last two sections are not experiments - they audit what this repository may lawfully publish, and what it actually exposed when it was published. Each section names the report file it came from.

Two conventions run throughout. Intervals are 95% and **verdicts are read from the pessimistic end**, never the point estimate. Rates whose sample could be all-correct use Clopper-Pearson intervals rather than a bootstrap, because a bootstrap collapses to [1.00, 1.00] on an error-free sample and would report an artifact of the estimator as a result.

Summary of every test and audit

#	Test	What it answers	Headline	Verdict
1	Engine screening (Phase 3)	Which engine should be the product	MACE-MPA-0: F1 0.852, precision 0.853, MAE 28 meV/atom	Adopted
2	Validation scorecard (calibration)	Accuracy and routing on 3,998 WBM candidates	Precision 0.826, NPV 0.971, 43% to DFT	Use with caution
3	Locked WBM test (one-time)	Do the frozen rules hold out of sample	Precision 0.960, NPV 0.988, 41% to DFT	Passed
4	Unseen real materials	Does it work on chemistry never trained on	Precision 1.00 [0.85, 1.00] on 22 calls; 82% to DFT	Not distinguishable from WBM
5	Polymorph ranking	Which form of a composition wins	Ground state first 0.69; gap MAE 19.4 meV/atom	Use with caution
6	Cost operating point	Where to set the threshold given costs	-35 meV/atom; precision 0.94, recall 0.45	Policy-derived, not measured
7	Oxide/halide calibration (new)	Can these families be certified at all	Unstable side certified for both; stable side neither	Split result
8	Energy-plausibility guard	Was a bad result corrupting the statistics	One impossible energy (-1131 eV/atom) found and excluded	Fixed
9	Leakage check (SevenNet-Omni)	Is a rival engine's win trustworthy	~1% of training undocumented	Not adopted
10	Reference data collection (new)	Citable experimental data for future suites	89 rows across 2 properties; experimental floor ~0.09%	Delivered
11	Licensing and attribution audit	What may lawfully be published, and under what	One real gap (MP attribution); ASE is the one LGPL dependency	Gaps closed
12	Pre-publication audit	Does anything private reach a public repository	0 secrets in history; a personal email reached the public repo	Closed, with one real failure

1. Engine screening - choosing the production model

Source: reports/phase3/candidates.md, screening.md. Paired comparison on a fixed subset (125 MP pairs per hull bin plus 500 WBM calibration structures). The locked test set was not used.

The **F1**, **DAF**, **precision** and **MAE** columns below are the engines' *published* Matbench Discovery scores, read from the model metadata files, not measurements made by this harness. This harness's own measurement of the adopted engine is F1 0.832 [0.808, 0.853] (section 2), which contains the published 0.852 - reproducing the benchmark is itself one of the results. The two numbers are different quantities and should not be compared as if one contradicted the other.

Engine	F1	DAF	Precision	MAE meV/atom	Compliant	Speed on this Mac
MACE-MP-0 medium	0.669	3.777	0.577	57	yes	8.6 s/relaxation
MACE-MPA-0 medium	0.852	5.582	0.853	28	yes	9.8 s/relaxation
eSEN-30M-OAM	0.925	6.069	0.928	18	yes	92.3 s/relaxation
SevenNet-Omni	0.906	5.954	0.910	21	unverified	73.6 s/relaxation

Decision: MACE-MPA-0 medium. eSEN-30M-OAM scores better on every accuracy metric but runs roughly 9x slower, which puts a full run at about a week on this machine and makes it unusable here. SevenNet-Omni is faster than eSEN but its training provenance is unverified (test 9). MACE-MP-0 is retained as a second engine, used only for the disagreement signal.

2. Validation scorecard - calibration set

Source: reports/validation_report.md. 3,998 usable WBM calibration relaxations. Every threshold in the product is fitted here and nowhere else.

Quantity	MACE-MP-0	MACE-MPA-0	Change
Precision at optimal threshold	0.686	0.826	+0.140
Recall at optimal threshold	0.534	0.838	+0.304
F1	0.600	0.832	+0.232
NPV	0.919	0.971	+0.052
Discovery acceleration factor	3.607	5.411	+1.804
Share routed to DFT	0.636	0.433	-0.203
Energy MAE, 0-0.025 bin	26.0	11.0	-15.0
Energy MAE, >0.3 bin	124.3	84.3	-40.0

Energy error grows sharply with distance from the hull: trustworthy below 0.1 eV/atom (11-19 meV/atom), but 97 meV/atom above 0.3 eV/atom, which is why the product refuses to rule on candidates in that range.

A second finding settled a design question: relaxing every competing phase with the engine (mode b) makes hull-distance error *worse* by 1.9 [0.8, 3.2] meV/atom than placing the engine's energy on the DFT hull (mode a). On new materials the error belongs to the new structure and does not cancel against competitors. Mode (a) is used, and it needs no competitor relaxations at all.

3. Locked WBM test - the one-time evaluation

Source: reports/final_test.md. Opened once on 2026-09-12. 3,999 usable relaxations, 1 rejected by the guard. Every rule was fitted on the calibration set before this set was touched.

Routing quantity	Value
n	3,999
likely stable	224
likely unstable	2,146
send to DFT	1,629
precision of 'likely stable'	0.960 [0.93, 0.98] - target 0.90
NPV of 'likely unstable'	0.988 [0.984, 0.993] - target 0.95
share sent to DFT	0.407
truly stable found as 'likely stable'	0.352
truly stable wrongly called 'likely unstable'	0.041

Both targets were met out of sample, and precision came in above its calibration value rather than below it. This is the single strongest result in the project.

Energy error by hull bin (test set)

Hull bin (eV/atom)	n	Energy MAE	median [err]	trimmed mean
<0	611	15.3 [13.0, 18.5]	8.1	10.2
0-0.025	416	11.0 [9.7, 12.5]	6.7	8.1
0.025-0.1	1,120	14.1 [13.3, 15.1]	9.3	11.3
0.1-0.3	1,334	30.2 [27.6, 32.9]	16.4	20.0
>0.3	518	86.7 [73.0, 103.1]	32.9	51.5

4. Unseen real materials - the hardest test

Source: reports/unseen_test.md. 325 real Materials Project materials that cannot be in the engine's MPtrj training data, each reached the way a user reaches a candidate - a known parent prototype plus one element substitution - and scored end to end through the product, never through a suite.

Quantity	Locked WBM test	This unseen set	Reweight to WBM bin mix
precision of 'likely stable'	0.96 [0.93, 0.98] (215/224)	1.00 [0.85, 1.00] (22/22)	1.00 [1.00, 1.00]
NPV of 'likely unstable'	0.988 [0.983, 0.992] (2121/2146)	0.947 [0.823, 0.994] (36/38)	0.980 [0.945, 1.000]
share sent to DFT	0.407	0.815	0.793 [0.729, 0.851]
prevalence of truly stable	0.153	0.274	0.153

Energy accuracy survives on real chemistry - 7.1 meV/atom MAE in the below-hull bin where the structure was found correctly. **The decision rates are not distinguishable from WBM's**, but only because 22 and 38 calls cannot tell them apart; this is an absence of evidence, not evidence of equivalence.

The finding that matters most, though, is about coverage rather than accuracy: the share of candidates sent to DFT doubled, from 41% to 82%.

Why the DFT share doubled - the central structural finding of this project. Certified thresholds are per chemistry family, and a family with no certified threshold can never receive that label whatever the engine predicts. The locked WBM test set is 53% f-electron, and f-electron was the *only* family in which a 'likely stable' threshold could be certified at all. What Materials Project actually adds is complex oxides and halides - 48% and 12% of this set, against 5% and 5% of WBM - and neither family had any certified threshold. **The published precision was measured almost entirely on a chemistry that barely appears in real use.** Test 7 is the direct response to this.

Structure-finding rate - measured here for the first time

A product outcome rather than a diagnostic: if the relaxation lands somewhere else, every number attached to it describes a material the user did not ask about. Nothing in the WBM numbers speaks to this, because a WBM candidate starts from WBM's own initial structure.

Hull bin	n	found the target structure	relaxation left its start
<0	89	0.89 [0.82, 0.96]	11%
0-0.025	83	0.86 [0.78, 0.93]	16%
0.025-0.1	105	0.74 [0.67, 0.83]	26%
0.1-0.3	32	0.94 [0.84, 1.00]	6%
>0.3	16	0.38 [0.12, 0.62]	50%
all	325	0.81 [0.77, 0.85]	18%

The 0.38 rate above 0.3 eV/atom is the real justification for the product's refusal to rule in that range - not the energy error, which had been the stated reason.

5. Polymorph ranking

Source: reports/polymorph.md. 297 Materials Project compositions with 3-8 known polymorphs inside a 0.2 eV/atom window. A scientist rarely asks whether one exact structure is stable - they ask which form of a composition wins.

Quantity	Value	Verdict
ground state ranked first	0.69 [0.63, 0.74]	use with caution
Spearman rho of the full ordering	0.66 [0.60, 0.71]	use with caution
error on energy gaps from the ground state	19.4 [17.3, 21.7] meV/atom	trustworthy
... median error	8.2 [7.2, 9.0] meV/atom	-

Random guessing would be right about 26% of the time, so 0.69 is real skill. But the result is strongly conditional on difficulty: where the DFT gap to the runner-up is under 10 meV/atom, the ground state is picked first only 0.55 [0.47, 0.62] of the time - close to a coin flip, and below the engine's own error on new materials. Above a 50 meV/atom gap it reaches 0.87.

In 14% of compositions the relaxation drove two distinct forms into the same structure. Their relative ranking is meaningless whatever the energies say, and a product that reports an ordering without disclosing this is reporting noise as a result. Also: these are known materials the engine trained on, so this measures ranking skill on material it has already seen.

6. Cost-based operating point

Source: reports/costs.md. Computed on the 3,998 usable WBM calibration structures; the locked test set is never re-optimised against.

Threshold -35 meV/atom: precision 0.94 [0.91, 0.97], recall 0.45 [0.42, 0.49], NPV 0.910, DAF 6.18. 294 of 3,998 candidates called stable, 17 of them wrong, and 333 stable materials missed.

These three cost coefficients are not the same kind of number, and the report is explicit about it. A wasted lab test = 1 is a normalisation baseline, fixed by definition. A missed stable material = 0.11 is *policy-derived, not empirical*: it is the ratio implied by the already-frozen 90% precision target, $(1-0.9)/0.9$, not a measurement of what a missed material costs. $\text{cost_dft} = 0.02$ is an order-of-magnitude estimate whose denominator - what one synthesis and characterisation attempt actually costs - has no defensible source in this project. Conclusions depending on it should be reported across 0.005-0.1. The sensitivity table, not the point estimate, is the trustworthy part of that page.

Read the other way, the 90% precision target is itself a cost claim: it assumes a wasted lab test is nine times worse than missing a stable material. If that is not the view, the target should move, not just the threshold.

7. Oxide and halide calibration - new, 17 September 2026

Run in this session. A new calibration task built to answer the coverage gap that test 4 exposed: 4,000 oxide and 3,500 halide calibration candidates, drawn only from pool ids the original split never used, with 8,000 prior ids excluded and 655 candidates dropped for sharing a reduced formula with a locked-test id. 15,000 jobs, 0 failures.

Before this run, oxide (n=185) and halide (n=143) certified **nothing** on either side - every candidate in roughly 60% of real MP additions went to DFT by default. The question was whether more data would fix that.

Family	n usable	base rate	stable side	unstable side
oxide	3,994	0.112	not certified	+0 meV NPV 0.9781, CP-lower 0.9692
halide	3,500	0.232	not certified	+0 meV NPV 0.9644, CP-lower 0.9516

The unstable side now certifies for both families. Routing impact: 88.1% of oxide candidates and 74.7% of halide candidates can be discarded without DFT, against roughly 100% going to DFT before.

The stable side failed for both - but for two different reasons, and the distinction decides what to do next:

Family	Best point precision	CP-lower	Diagnosis	What would fix it
halide	0.9401 (t=-70 meV)	0.8803	Sample-size limited. True precision is already above the 0.90 target; only the confidence bound falls short.	~1.8x more halide calibration structures (~6,300 vs 3,500). The pool holds 10,095.
oxide	0.8798 (t=-30 meV)	0.7938	Precision limited. Point precision peaks below target, so the bound converges to something under 0.90 regardless of n.	Nothing, at 0.90. It certifies at a 0.80 target (t=-20 meV); 0.85 does not certify.

A satisfied NPV target is not the same as keeping your discoveries. Both families clear NPV 0.95, but because the base rates are low, discarding 88.1% of oxide candidates still loses **17.3% of all truly stable oxides** (77 of 446); halide loses 11.4% (93 of 813). Both numbers should be quoted whenever this threshold is proposed for routing.

A methodological caution came out of this run

The calibration was first run on the wrong engine - MACE-MP-0 rather than the production MACE-MPA-0 - because HARNESS_MODEL defaults to the former when unset, and the run still reported zero failures. On that weaker engine, halide's precision ceiling looked like 0.8313 and the conclusion would have been that *neither* family could ever be certified. The production engine is far more accurate on this chemistry (halide MAE 27.9 vs 59.9 meV/atom; over-stabilisation -10.9 vs -44.8), and that moved halide's ceiling above target. **A certification verdict cannot be generalised across engines.**

8. Energy-plausibility guard

Source: reports/guard_fix.md.

One converged relaxation with a physically impossible energy - **-1131.01 eV/atom** against an MP reference of -4.09 - was being averaged into the reported statistics. It had converged cleanly: max stress 0.0096 GPa inside the 0.01 limit, final fmax 0.0018 eV/A inside the 0.01 limit, and it even matched the expected structure. The relaxation had fallen into a spurious short-range minimum of the model's potential-energy surface, just wide enough to keep the nearest-neighbour ratio above the collapse cut at 0.514. Neither existing guard - convergence, or minimum interatomic distance - could see it, and a single row dominated every mean it entered. Two more results in the same class were found by sweeping the whole database.

Nothing was loosened to produce the corrected numbers: no settings tag changed, no tolerance moved, no hard case was dropped. The guard only ever moves a result from *scored* to *counted-and-excluded*.

9. Training-data leakage check - SevenNet-Omni

Source: reports/leakage_check.md. SevenNet-Omni outscored the adopted engine on the screening subset; this asks whether that win is real or an artifact of WBM structures being in its training data.

Portion of training data	WBM-filtered?	Evidence
OMat24 (~46%)	yes	dataset authors filtered on WBM prototype labels and say so
sAlex (3D Alexandria)	yes	same
Alexandria 1D/2D, MatPES, MAD (~1%)	not documented	a 3D WBM prototype in a 1D/2D trajectory is structurally unlikely, but unverified

Verdict: the win is probably real, but the engine was not adopted. Its registry status stays 'unverified' - no longer unexamined, but honest. Settling it needs no relaxations and no GPU: compute AFLOW protostructure labels for the 8,000 WBM ids in the split and intersect them with the undocumented training portions.

A definitional trap worth knowing. 'Compliant' in this project means training data verified free of WBM, which is this project's definition and not Matbench Discovery's. By the leaderboard's stricter rule - train on the sanctioned set only - MACE-MPA-0 and eSEN-30M-OAM are non-compliant too, despite being marked True here. That column must not be read as a leaderboard compliance flag.

10. Reference data collection - new, 17 September 2026

Source: reference_data/COVERAGE.md, SOURCES.md. Citable experimental data for property suites to be validated later. Gathering and verification only - no simulation was run and no existing suite was modified.

Property	Rows	Materials	Source	Licence
Oxide lattice constants (RT)	48	22 formulae 30 phases	Crystallography Open Database	CC0 - commercial OK
Atomic oxygen erosion	41	39 polymers	NASA/TM-2006-214482 (MISSE 2 PEACE)	US Gov - public domain
Outgassing (TML/CVCM)	0	-	NASA database - blocked by CAPTCHA; RP-1124 scans OCR to noise	-

The cross-check result is the useful output. Where two independent experiments of the same phase disagree by x, chasing simulation accuracy below x is not measurable. Across 8 oxide phases with two or more independent determinations, the spread in the lattice parameter is a **median 926 ppm (0.093%) and at worst 1,991 ppm (0.199%)**. So a simulation targeting 1% sits comfortably above the experimental floor, and targeting better than about 0.1% is chasing noise between experiments.

Two data-integrity findings came out of building it:

The Materials Project was deliberately excluded as a source. Its elasticity data is DFT-computed, and the production engine is trained on MP data, so using it as a reference would measure agreement with the engine's own training distribution rather than with reality. Independent experimental values are the only ones that add information.

COD needs curation, not just querying. Raw formula matching returned 100 rows, of which 52 were contaminated. COD's structured pressure field is null even for explicit diamond-anvil studies - one ZnO series records a cell temperature and no pressure at all, while its title says 'High-pressure X-ray investigation'. Compression and in-situ heating series masquerade as repeat measurements, formula matching pulls in unrelated compounds (one SiO₂ hit was a zeolite), and doped samples sit under the parent formula - the first Al₂O₃ hit is ruby.

Licence position: the generated tables, schemas and prose are CC BY 4.0 (`reference_data/LICENSE`); the loader and extractors are Apache-2.0 with the rest of the project's code. Two public-domain sources are vendored verbatim under `reference_data/raw/` so the build reproduces offline - the NASA memorandum and the cached COD responses - and both are excluded from this project's grants, keeping their own status.

Licensing trap, verified at source. The NIST Chemistry WebBook is Standard Reference Database 69 and is *copyrighted* - 'All rights reserved', secured under 15 U.S.C. 290e - even though the numbers are free to read, and NIST reserves the right to charge for access in future. Individual facts are not copyrightable but the compilation is, so from any NIST SRD product this collection stores a small number of cited values and never a bulk table. No NIST SRD data is used in any shipped table.

11. Licensing and attribution audit

Source: `reports/licensing.md`. What this repository uses, what it redistributes, what each licence requires, and whether it complies. Written for publication review. Not legal advice; close judgements are flagged as close.

What is actually redistributed

'Redistributes' means a file in the repository contains a third party's values, not just code that fetches them. That distinction does most of the work here.

Asset	Licence	Redistributed?	Position
Materials Project - ids, formulas, hull distances, PBE energies	CC BY 4.0	yes - derived values throughout data/ and reports/	Adaptations, so attribution required
WBM / Matbench Discovery	CC BY 4.0	ids only - structures and energies are gitignored	Fine; the reader downloads from figshare
MPtrj (engine training set)	MIT	no - never downloaded	Cited only
Experimental lattice constants (40 + 29 rows)	journal copyright	yes - transcribed with citation	The closest call in the document
Model checkpoints	MIT, except eSEN (OMat24, gated)	no - URL + SHA-256 only	Must stay that way

The experimental tables are the one close call. Individual measured values are facts and are not copyrightable, but copyright can reach the *selection and arrangement* of a table, and two specific tables are reproduced substantially. The assessment is low risk and standard scientific practice - these are benchmark reference sets that exist to be reused, and both files carry the full citation. Reducing the risk to zero would mean shipping a script that reconstructs them from papers the reader fetches, which costs reproducibility for anyone without journal access. Deliberately not done.

Dependencies - one matters commercially

Every dependency is permissive **except ASE, which is LGPL-2.1-or-later**. LGPL's obligations attach to *distributing* the library, which this repository does not do: it declares a dependency and the user's package manager fetches it. Code that imports ASE is not made copyleft by that. The obligation bites on shipping a bundle - a PyInstaller binary, a Docker image, a vendored wheel - at which point the recipient must be able to replace their copy of ASE. Nothing to do for this repository; it is the one dependency to plan around for a commercial product.

Gaps found, and their current state

Requirement	At audit	Verified 17 Sep 2026
Materials Project attribution and licence notice	MISSING - the only real gap	Closed - NOTICE and README block present
WBM / Matbench Discovery attribution	missing (same fix)	Closed
A licence for this project's own code	none	Closed - Apache-2.0 at root
Experimental tables: citation	present	unchanged
Checkpoints not redistributed; gated eSEN respected	yes	unchanged
Nothing third-party vendored	yes	unchanged

The report offered five licensing options and recommended **Apache-2.0 for the code and CC BY 4.0 for reports/ and data/** - matching the inputs, giving corporate users the patent grant that bare MIT does not, and keeping the inspectability that publishing is for. That recommendation was adopted: LICENSE (Apache-2.0),

NOTICE with the licence split, and data/LICENSE and reports/LICENSE (CC BY 4.0) are all present.

A permissive licence is not a moat, and the report says so plainly. The code is a harness whose entire value is that the numbers can be checked, so a licence that prevents checking removes the reason to publish. No dependency is copyleft-in-distribution and no input dataset requires share-alike, so nothing forced the choice - Apache-2.0 does not stop a competitor using the harness, and no permissive licence would.

12. Pre-publication audit

Source: reports/publish_audit.md. Audit of the repository before it was made public. **At audit** (commit 82a77c1): 28 commits across 2 branches, 127 tracked files - those counts are left unedited in the source report as the record of what was actually scanned. **As published** it has 30 commits and 141 tracked files, and the commit author email has been rewritten.

Credential sweep - scope, then result

The scope is the point. The sweep covered all 28 commits on all branches, **all 437 objects in the object database** including 277 blobs - a superset of 'every file in every commit', because it includes objects no ref points at - 0 dangling blobs, the relog (58 entries, all already on refs), and 34,035 working-tree files including caches.

Check	Result
MP_API_KEY value anywhere in git history	0 blobs (searched by exact value, all 277)
MP_API_KEY value in the working tree	1 file: .env - gitignored, never tracked
.env (any flavour) in history	no - only .env.example, empty placeholders
Hugging Face token	none - the gated checkpoint left no token behind
Private keys, cloud creds, PATs, JWTs, DB strings	none
URLs with embedded credentials	none

gitleaks reported exactly one history finding and it is a false positive: the string MP2020Compatibility in a code comment, long and mixed-case enough to trip a generic entropy rule. Roughly 40 further working-tree hits are all inside gitignored third-party cache.

The audit also disclosed a mistake it made itself: its own pattern matcher echoed a 13-character prefix of the live API key into the session transcript. 19 of 32 characters remained unknown, so the risk was low - and it recommended rotating the key anyway, on the grounds that the value of an audit is not having to reason about how low 'low' is.

Honesty of the public claims

Stale reports were stamped rather than quietly deleted: a 'not the production engine' banner on the baseline scorecard, a 'superseded' banner on the pre-guard Phase 0 report, and a placeholder-costs warning on the cost page. **All three banners are emitted by the generators, not pasted into the files**, so regenerating a report cannot silently drop them. Every README figure is a pessimistic bound linked to the section it came from.

The finding that matters most - and it is a failure

A personal email address reached the published repository. The repository was pushed and made public on 2026-09-14. About twenty minutes later an independent re-verification found the owner's personal address in the published *history* - two occurrences inside the audit document itself, in the commit where it quoted the address while explaining the email decision. A later commit reworded those lines, so the tip was clean and **every check in the chain was blind to it**: the verification step read commit metadata, which was correct; the preflight used git grep, which reads tracked content at HEAD, where the address no longer appeared; and gitleaks does not flag email addresses at all, so its green tick was accurate and irrelevant. Nothing in the chain scanned historical file content. That is the defect.

Response. The repository was set private within an exposure window of roughly 20 minutes, with 0 forks, 0 stars and 0 watchers. A second filter-branch pass redacted the string across history and commit messages, verified to 0 matches across all 513 objects reachable from both branches, with the HEAD tree hash and the full 142-file manifest unchanged - only the one historical commit's tree moved. The remote was replaced by delete-and-recreate rather than force-push, because GitHub keeps orphaned commits reachable by SHA after a force-push.

Fix. The preflight script now scans every blob, tree and commit reachable from HEAD, the branches and the tags - file content *and* commit metadata - rather than the working tree alone. It was regression-tested against the pre-redaction history: the old check reports clean, the new one names the offending blob and file and exits 1.

The general lesson, worth more than the specific fix: a check that reads HEAD cannot clear a history. Anything published as history has to be audited as history.

Left open deliberately

Item	Status
Install gitleaks locally	not done - CI covers it
Drop data/auto_pairs_v1.json (1.31 MB, superseded)	kept - results.parquet holds results for 1,982 v1-only pairs; deleting it would orphan them
Replace the two experimental CSVs with a reconstruct-from-paper script	not done - costs reproducibility, assessed low risk
Expire refs/original/	kept as the in-repo rollback path; not published by an ordinary push

What the tests say when read together

1. The product is accurate where it matters, and honest about where it is not

Energy error is trustworthy below 0.1 eV/atom from the hull (7-14 meV/atom on real unseen chemistry) and untrustworthy above 0.3 eV/atom (87-291 meV/atom). Both decision targets were met out of sample on the locked test set, with precision 0.960 against a 0.90 target.

2. Coverage, not accuracy, is the binding constraint

The single most consequential finding across all ten tests is that the headline precision was measured on a chemistry that barely appears in real use. The locked WBM test is 53% f-electron and f-electron was the only family that could certify a stable threshold; what Materials Project actually adds is oxides and halides. On real unseen materials the DFT share doubled to 82% for exactly this reason. Test 7 addressed it directly and got a split answer: halide is fixable with about 1.8x more calibration data, oxide is not fixable at a 0.90 target at all.

3. Structure-finding is a distinct failure mode, and it was invisible until recently

The pipeline returns the material actually asked for 0.81 [0.77, 0.85] of the time, and only 0.38 [0.12, 0.62] of the time above 0.3 eV/atom. No WBM-derived number speaks to this, because a WBM candidate starts from WBM's own structure. Where the relaxation lands elsewhere, energy error is roughly 20x higher (414.6 vs 24.9 meV/atom in the top bin).

4. Several headline numbers rest on assumptions that are not measurements

The cost coefficient for a missed stable material is derived from the precision target rather than measured, so the two cannot be used to justify each other. The compute-to-experiment cost ratio has no sourced denominator. 'Compliant' means something narrower here than on the Matbench Discovery leaderboard. Each of these is documented at its source, and none should be quoted as established.

5. Methodological guards that earned their place

Three habits caught real errors during this work. **Clopper-Pearson rather than bootstrap** on rates whose sample could be all-correct - a bootstrap was caught producing a false verdict on an error-free sample.

Pessimistic-end verdicts, which keep a favourable point estimate from overstating a thin sample. **Counting rejected results rather than dropping them** - the energy-plausibility guard moves a result from scored to counted-and-excluded, so exclusions can never quietly improve a mean.

6. The same discipline was applied to what gets published, and it still failed once

The publication audits are cut from the same cloth as the tests: define the scope explicitly, then report what was found rather than what was hoped for. The credential sweep covered every object in the database rather than every file at HEAD, and found nothing. The licensing review named one real gap - missing Materials Project attribution - rather than declaring compliance, and that gap is now closed under Apache-2.0 and CC BY 4.0.

And it still missed something: a personal email address reached the public repository inside history, because every check in the chain read HEAD or commit metadata and none read historical file content. The exposure was twenty minutes with no forks. The failure is recorded in the audit document in full, alongside the fix, rather than being quietly corrected - which is the same convention the test reports follow when a result is unflattering.

Locked and unopened as of this document: the original WBM locked test set (4,000 ids) has been opened once, on 2026-09-12. The two per-family locked test halves created on 2026-09-17 - 2,000 oxide and 2,000 halide - remain closed, and no code path in this session passed unlock=True.